

Families of Classification Boundaries:

Locally Adaptive Support Vector Machines Against Matched Global References,
with an Exact Statement of When Local Adaptation Cannot Help

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Abstract

We revisit a paused research line on locally adaptive support vector machines, reconstructed (the original variants are not available) as four small families—kNN-localised, cell-partitioned and soft-mixture local linear SVMs, and an RBF-SVM with a positive-definite variable-bandwidth kernel—each containing a global member as a special case. First, an exact statement of when local adaptation cannot help: with cells chosen independently of the labels, the expected risk is the reference’s learning curve averaged at binomially thinned sample sizes, never smaller if that curve is non-increasing and strictly larger if it still decreases at the last step; in a Gaussian model with $n = 320$, four cells multiply the excess risk of the nearest-centroid threshold by 4.08 (closed form). Second, a comparison on 6 data sets under a criterion of added value fixed before running. No family met it: of 24 data-set/family intervals, 9 lie below zero and 1 above (about 1.2 expected above zero by chance). In the multiscale regime the variable-bandwidth family ties the RBF-SVM and the local linear families stay 3.5–7.6 points below it in 5 of 6 cells; both outcomes are grid-limited, as the inner cross-validation still selects grid edges there. Under label noise and subsampling 4 of 48 intervals lie above zero (up to about 4.0 expected by chance) and kNN-SVM is below the reference on 3 and 3 of 6 data sets; no superiority in accuracy or robustness is claimed.

1 Introduction

A support vector machine with a fixed kernel draws one boundary for the whole input space [6, 12, 17, 19]. Several proposals let the boundary adapt to where the query point lies: train the classifier only on the neighbours of the query [3, 5, 18, 24], split the space into cells with one model each, let the kernel width follow the local density as in variable-bandwidth density estimation [1, 21], or blend local linear classifiers with soft weights [13]. Each of these can, in principle, fit a boundary whose curvature or scale varies across the space; each also fits its local model on fewer points than the global reference.

The research line SVMF001 studied such “families of classification boundaries” and recorded, in its public summary, that the exploratory improvements depended strongly on the regime and that the final comparison did not support the required added value over comparable references, with no claim of general superiority in robustness or classification. The concrete variants and the regimes are not available to us. This draft therefore does two things, in the same spirit and at a deliberately modest scale: it states and proves, in a precise but limited form, *when local adaptation cannot help* (Section 3), and it reconstructs the comparison as a controlled experiment with a criterion of added value fixed in advance (Sections 4–5).

Scope. The families are reconstructions, not the original variants; the data sets are small (at most 400 points, for a budget of a few CPU-minutes); the exact statement covers partition-based localisation and not neighbourhood-based localisation, which is only illustrated. Section 6 lists every claim with its status. Version v0.2 applies one round of internal review: the hyperparameter grids of v0.1 were found to be saturated by the inner cross-validation on the data sets that decide the result, so they were widened and the comparison was re-run (Section 4).

2 Setting and the four families

Let (X, Y) be a random pair in $\mathcal{X} \times \{-1, +1\}$, $\mathcal{X} \subseteq \mathbb{R}^d$, with distribution P , $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$. The *risk* of a classifier $f: \mathcal{X} \rightarrow \{-1, +1\}$ is $R(f) = \mathbb{P}(f(X) \neq Y)$, the *Bayes rule* is $f^*(x) = \text{sign}(2\eta(x) - 1)$ with risk R^* , and the *excess risk* is $\mathcal{E}(f) = R(f) - R^* \geq 0$ [9]. A *learning rule* A maps a finite labelled sample S of any size $m \geq 0$ to a classifier $A(S)$ (with a fixed default for $m = 0$). Its *learning curve* is $R_A(m) = \mathbb{E} R(A(S_m))$ for S_m an i.i.d. sample of size m from P .

Definition 2.1 (Approximation and estimation error). For a class $\mathcal{F}$ of classifiers and $\hat{f} \in \mathcal{F}$ fitted on a sample,

$$\mathcal{E}(\hat{f}) = \underbrace{\inf_{f \in \mathcal{F}} R(f) - R^*}_{\text{approximation}(\mathcal{F})} + \underbrace{R(\hat{f}) - \inf_{f \in \mathcal{F}} R(f)}_{\text{estimation}(\hat{f})}.$$

Both terms are non-negative; the approximation term is zero if and only if $\mathcal{F}$ contains a Bayes rule (up to null sets).

Global references and locally adaptive families. All families are fitted on standardised inputs. The two references are the soft-margin *linear SVM* $f(x) = \text{sign}(\langle w, x \rangle + b)$ and the *RBF-SVM* with kernel $k_\gamma(x, z) = \exp(-\gamma\|x - z\|^2)$ [4, 6, 22], both fitted with the hinge loss, which is classification-calibrated [2, 25], and both with the penalty parameter C and, for the RBF, γ chosen by inner cross-validation from the grids of Section 4. The four locally adaptive families are:

- (a) **kNN-SVM** (localised linear SVM) [3, 24]. For a query x let $N_k(x)$ be the k training points nearest to x . If all their labels coincide, predict that label; otherwise predict with a linear SVM (fixed C) trained on $N_k(x)$ alone. The value $k = \text{“all”}$ gives the global linear SVM with the same C .
- (b) **Cell-SVM** (hard partition). Partition the training inputs into M cells by k -means; fit one linear SVM per cell (a constant vote if the cell is pure); route a query to the cell of its nearest centre. $M = 1$ is the global linear SVM.
- (c) **VB-RBF-SVM** (variable-bandwidth kernel). Give each point x a bandwidth $a(x) = a_0 (r_\kappa(x)/\bar{r})^{2\beta}$, where $r_\kappa(x)$ is the distance from x to its κ -th nearest training point (excluding itself when x is a training point), $\bar{r}$ is the median of r_κ over the training points, $a_0 = 1/(2\gamma)$ and $\beta \geq 0$; then use the kernel K_a of Lemma 2.2 in an SVM. $\beta = 0$ is the RBF-SVM with parameter γ ; $\beta = \frac{1}{2}$ makes the bandwidth proportional to the κ -NN radius, the classical square-root-of-density scaling in spirit [1].
- (d) **LLSVM-style mixture** of local linear SVMs, a simplification of the locally linear SVM of Ladický and Torr [13] in which the local coding is replaced by softmax weights of the distance to anchors. With anchors $c_1, \dots, c_M$ from k -means and weights $w_m(x) \propto \exp(-\|x - c_m\|^2/2s^2)$ summing to one (s^2 the mean squared distance from the training points to their nearest anchor), the decision function is $f(x) = \sum_m w_m(x) (\langle \beta_m, x \rangle + b_m)$, which is linear in the feature map $\Phi(x) = (w_m(x)x, w_m(x))_{m \leq M}$ and is therefore fitted as one linear SVM on $\Phi(x)$. $M = 1$ is the global linear SVM.

Lemma 2.2 (A positive-definite variable-bandwidth kernel). *[proved here, standard argument] Let $a: \mathbb{R}^d \rightarrow (0, \infty)$ be any function. Then*

$$K_a(x, z) = \left(\frac{2\sqrt{a(x)a(z)}}{a(x)+a(z)} \right)^{d/2} \exp\left(-\frac{\|x-z\|^2}{a(x)+a(z)}\right)$$

is a positive-definite kernel on $\mathbb{R}^d$ in the sense of Aronszajn (all Gram matrices are positive semidefinite) with $K_a(x, x) = 1$, and for constant $a \equiv 1/(2\gamma)$ it equals k_γ .

Proof. Put $\varphi_x(u) = \exp(-\|u - x\|^2/a(x)) \in L^2(\mathbb{R}^d)$. Completing the square, $\|u - x\|^2/a + \|u - z\|^2/b = (1/a + 1/b)\|u - c\|^2 + \|x - z\|^2/(a + b)$ with $c = (bx + az)/(a + b)$, so $\langle \varphi_x, \varphi_z \rangle = (\pi ab/(a + b))^{d/2} \exp(-\|x - z\|^2/(a + b))$ with $a = a(x)$, $b = a(z)$, and $\|\varphi_x\|^2 = (\pi a/2)^{d/2}$. Hence $K_a(x, z) = \langle \varphi_x, \varphi_z \rangle / (\|\varphi_x\| \|\varphi_z\|)$, a normalised Gram function of vectors in a Hilbert space, whose Gram matrices are positive semidefinite. The last claim is the case $a = b = 1/(2\gamma)$. $\square$

Remark 2.3 (Nesting, and what it does and does not guarantee). Every local family contains a global member in its own grid ($k = \text{all}$, $M = 1$, $\beta = 0$), and in the experiment ties in the inner cross-validation are broken towards that member. For VB-RBF-SVM the member is the RBF-SVM over the same grid as the reference; for cell-SVM and LLSVM it is a linear SVM tuned over a sub-grid of the reference’s C values ($C \geq 0.1$ and $C \geq 0.01$ against $C \geq 0.001$; LLSVM with a regularised constant feature); for kNN-SVM it is the linear SVM with $C = 1$ fixed. In population risk a local family can therefore fall below its own global member only through selection noise and rise above it only through genuine adaptation; observed test-fold differences also carry test-sample noise in both directions. It can rise above the *selected* best global reference also through noise in the selection of that reference (the inner cross-validation may pick the worse of the two references on the test fold), which is what Section 5 finds; the nested comparison against the fixed member is therefore reported as well (Table 7). The kernel prefactor in Lemma 2.2 is what makes K_a a valid kernel; in dimension d it down-weights pairs with mismatched bandwidths by $(2\sqrt{ab}/(a + b))^{d/2}$, a strong penalty when d is large. Dropping it is not an option: the matrix $[\exp(-\|x_i - x_j\|^2/(a_i + a_j))]$ is not positive semidefinite in general; for 5 points on the line with bandwidths between 0.066 and 153 (listed in `results/exact_example.json`) its smallest eigenvalue is -0.96 , while that of K_a is 0.04.

3 When local adaptation cannot help

Lemma 3.1 (Only the estimation term is available). *[proved here, elementary] Let $\mathcal{F}_{\text{ref}}$ contain a Bayes rule. For any classifier $\hat{g}$ (local or not), $\mathcal{E}(\hat{g}) \geq 0 = \text{approximation}(\mathcal{F}_{\text{ref}})$. Consequently a locally adaptive procedure can improve on a reference fitted in $\mathcal{F}_{\text{ref}}$ only by having a smaller estimation term, never by reducing the approximation term, which is already zero.*

Proof. Immediate from Definition 2.1 and $\mathcal{E} \geq 0$. $\square$

The lemma is trivial but it locates the question: when the Bayes boundary is a hyperplane and the reference is a linear SVM, or more generally when the reference class is rich enough (a universal kernel [14]; see also Steinwart and Christmann [19, §4.6]), local adaptation is a bet that fitting on fewer points estimates better. The next proposition makes this exact for partition-based localisation.

Proposition 3.2 (Partition-localised rules with label-independent cells). *[proved here] Let A be any learning rule with learning curve R_A . Let $(X_i, Y_i, Z_i)_{i \leq n}$ and (X, Y, Z) be i.i.d., where $Z \in \{1, \dots, M\}$ is a cell label with $p_j = \mathbb{P}(Z = j) > 0$ and Z is independent of (X, Y) .*

The partition-localised rule predicts at a test point with cell $Z = j$ using $A(S^{(j)})$, where $S^{(j)} = \{(X_i, Y_i) : Z_i = j\}$. Then

$$\mathbb{E} R(\text{localised rule}) = \sum_{j=1}^M p_j \mathbb{E}[R_A(N_j)], \quad N_j \sim \text{Bin}(n, p_j).$$

If R_A is non-increasing, the right-hand side is at least $R_A(n)$. If moreover $R_A(n-1) > R_A(n)$ (in particular, if $n \geq 2$ and R_A is strictly decreasing on $\{1, \dots, n\}$) and $M \geq 2$, it is strictly larger than $R_A(n)$.

Proof. Given the training data with their cell labels, $\tilde{S} = (X_i, Y_i, Z_i)_{i \leq n}$ (on which every $S^{(j)}$ depends), the risk of the localised rule is $\mathbb{P}(A(S^{(Z)})(X) \neq Y \mid \tilde{S}) = \sum_j p_j \mathbb{P}(A(S^{(j)})(X) \neq Y \mid \tilde{S}, Z = j) = \sum_j p_j R(A(S^{(j)}))$, because (X, Y) is independent of Z and of $\tilde{S}$. Since the Z_i are independent of the (X_i, Y_i) , conditionally on $N_j = m$ the sub-sample $S^{(j)}$ is an i.i.d. sample of size m from P , so $\mathbb{E}[R(A(S^{(j)})) \mid N_j = m] = R_A(m)$, and $N_j \sim \text{Bin}(n, p_j)$. Taking expectations gives the identity. If R_A is non-increasing then $R_A(N_j) \geq R_A(n)$ because $N_j \leq n$. If $M \geq 2$ then $p_j < 1$ for every j , so $\mathbb{P}(N_j = n-1) = n p_j^{n-1} (1 - p_j) > 0$; on this event $R_A(N_j) = R_A(n-1) > R_A(n)$, and $R_A(N_j) \geq R_A(n)$ elsewhere, so $\mathbb{E} R_A(N_j) > R_A(n)$ for every j . $\square$

Remark 3.3 (What the proposition covers). (i) The cells may be defined through the inputs: if $X = (X_1, X_2)$ with X_2 independent of (X_1, Y) , A uses X_1 only, and Z is a function of X_2 , the hypotheses hold with X_1 in the role of X ; this is the case of a partition along directions that carry no label information. (ii) The identity holds for *any* rule A , including the SVM; the conclusion needs the monotonicity of R_A , which is a property of the pair (rule, distribution), not a theorem in general—learning curves can be non-monotone [23]. For the SVM in the model below it is checked numerically, not proved (Appendix B). (iii) Neighbourhood-based localisation (kNN-SVM) and data-dependent partitions (k -means cells, soft weights) are *not* covered: their local samples are not i.i.d. from P . Nothing here says that such rules cannot help in general; Lemma 3.1 says only that if they help, it must be through estimation.

Proposition 3.4 (Exact learning curve of the nearest-centroid threshold). *[proved here] Let Y be uniform on $\{-1, +1\}$ and $X \mid Y \sim N(Y\mu, 1)$ on $\mathbb{R}$, $\mu > 0$. The Bayes rule is $\text{sign}(x)$ with risk $R^* = \Phi(-\mu)$, and the rule “predict $+1$ iff $x > t$ ” has risk $R(t) = \frac{1}{2}[\Phi(t - \mu) + \Phi(-t - \mu)]$. Let A_{nc} fit $\hat{t} = \frac{1}{2}(\bar{X}_+ + \bar{X}_-)$, the midpoint of the class means, predicting the only present class when a class is missing. Then:*

- (i) *conditionally on the class counts (N_+, N_-) with $N_+, N_- \geq 1$, $\hat{t} \sim N(0, s^2)$ with $s^2 = \frac{1}{4}(1/N_+ + 1/N_-)$ and $\mathbb{E}[R(\hat{t}) \mid N_+, N_-] = \Phi(-\mu/\sqrt{1 + s^2})$; when a class is missing the risk is $\frac{1}{2}$;*
- (ii) *$R_{A_{\text{nc}}}(n) = \sum_{k=0}^n \binom{n}{k} 2^{-n} r(k, n-k)$, with $r(a, b) = \Phi(-\mu/\sqrt{1 + \frac{1}{4}(1/a + 1/b)})$ for $a, b \geq 1$ and $r(a, b) = \frac{1}{2}$ otherwise, satisfies $R_{A_{\text{nc}}}(0) = R_{A_{\text{nc}}}(1) = \frac{1}{2}$ and $R_{A_{\text{nc}}}(n+1) < R_{A_{\text{nc}}}(n)$ for every $n \geq 1$; in particular it is non-increasing on $n \geq 0$ and strictly decreasing on $n \geq 1$.*

Proof. (i) Given the counts, $\bar{X}_+ \sim N(\mu, 1/N_+)$ and $\bar{X}_- \sim N(-\mu, 1/N_-)$ are independent, so $\hat{t} \sim N(0, s^2)$. For $W \sim N(0, 1)$ independent of $\hat{t}$, $\mathbb{E} \Phi(\hat{t} - \mu) = \mathbb{P}(W - \hat{t} \leq -\mu) = \Phi(-\mu/\sqrt{1 + s^2})$ because $W - \hat{t} \sim N(0, 1 + s^2)$; by the symmetry $\hat{t} \mapsto -\hat{t}$ the second term has the same expectation. A constant rule has risk $\frac{1}{2}$ because the classes are equiprobable. (ii) The formula is (i) averaged over $N_+ \sim \text{Bin}(n, \frac{1}{2})$; with $n \leq 1$ a class is always missing, so the value is $\frac{1}{2}$. For monotonicity, realise the samples of sizes n and $n+1$ as the first n and $n+1$ terms of one i.i.d. sequence; the conditional expected risk given the labels is r of the counts in both cases, and the $(n+1)$ -th label raises one count by one. Since $r(a, b)$ is strictly decreasing in each argument on $\{a, b \geq 1\}$

M (cells)	expected risk	excess ($\times 10^{-4}$)	ratio to $M = 1$
1	0.05025	2.7	1.00
2	0.05052	5.3	2.01
4	0.05107	10.8	4.08
8	0.05221	22.3	8.38
16	0.05477	47.8	17.99
32	0.06638	163.9	61.68

Table 1: Exact values (finite binomial sums) for the partition-localised nearest-centroid threshold with M equiprobable label-independent cells in the Gaussian location model with $\mu = 1.645$, Bayes risk 0.0500, $n = 320$ (Corollary 3.5). The learning curve itself is strictly decreasing on $n = 2, \dots, 1000$ by direct evaluation, and for all $n \geq 1$ by Proposition 3.4; it is the right panel of Figure 1.

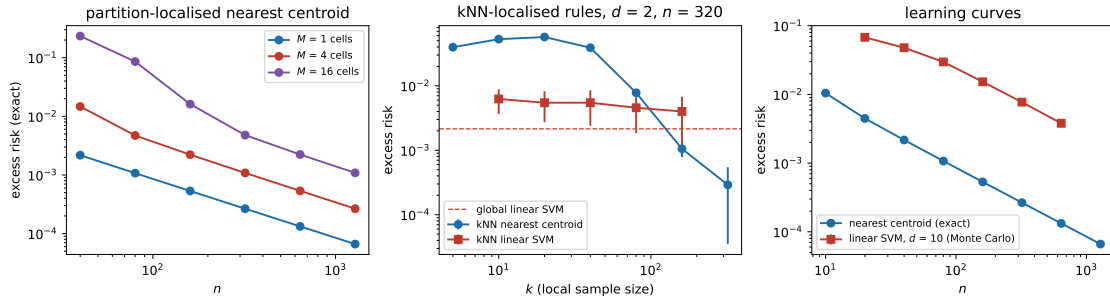


Figure 1: Left: exact excess risk of the partition-localised nearest-centroid rule against n for $M = 1, 4, 16$ label-independent cells (Corollary 3.5). Middle: kNN-localised rules in $d = 2$ (simulations D1, D2 of Appendix B); the dashed line is the global linear SVM. Right: learning curves of the nearest-centroid threshold (exact) and of the linear SVM in $d = 10$ (Monte Carlo, D3).

(s^2 decreases and $\Phi(-\mu/\sqrt{1+s^2})$ with it), $r(0, b) = r(0, b+1) = \frac{1}{2}$, and $r(a, b) < \frac{1}{2}$ whenever $a, b \geq 1$, the risk given the first $n+1$ labels is pointwise at most the risk given the first n , with strict inequality on the event that both classes are present among the first $n+1$ points, which has positive probability for $n \geq 1$. Taking expectations gives $R_{\text{Anc}}(n+1) < R_{\text{Anc}}(n)$ for $n \geq 1$. $\square$

Corollary 3.5 (An exactly computed case where local adaptation strictly hurts). *[proved here; numbers computed exactly] In the model of Proposition 3.4, extended by any coordinates independent of (X_1, Y) along which cells are formed, the partition-localised nearest-centroid rule with $M \geq 2$ equiprobable cells and $n \geq 2$ training points has strictly larger expected risk than the global one, and $\mathbb{E} R = \sum_m \binom{n}{m} M^{-m} (1 - M^{-1})^{n-m} R_{\text{Anc}}(m)$. (Proposition 3.2 applies because R_{Anc} is non-increasing and $R_{\text{Anc}}(n-1) > R_{\text{Anc}}(n)$ for $n \geq 2$; for $n = 1$ both rules have risk $\frac{1}{2}$.) With $\mu = 1.645$ (Bayes risk 0.0500) and $n = 320$: the global excess risk is 2.7×10^{-4} , and $M = 2, 4, 8, 16, 32$ cells multiply it by 2.01, 4.08, 8.38, 17.99 and 61.68 (Table 1). A Monte Carlo check with 2000 samples gives 0.05024 ± 0.00001 against the exact 0.05025 for the global rule and 0.05105 ± 0.00002 against 0.05107 for $M = 4$.*

Remark 3.6 (First-order reading). Since $R''(0) = \mu\varphi(\mu)$, the excess risk of a threshold with variance s^2 is $\frac{1}{2}\mu\varphi(\mu)s^2 + O(s^4)$, and $\mathbb{E} s^2 \approx 1/n$ for the global rule, $\approx M/n$ for a cell: the multiplication by about M in Table 1 is the variance inflation of fitting on n/M points, with no bias to be removed. The exact values are the finite sums, not this expansion.

Rules not covered by the proposition. Appendix B and Figure 1 report simulations in the same model for the rules the proposition does not cover (computationally verified, not proved): the risk of the kNN-localised nearest-centroid rule in $d = 2$ is at worst about twice the global

data set	regime	n	d	n_{train} (main)	n_{train} (sub25)	positives
wine	real	178	13	142	40	33%
breast cancer	real	300	30	240	60	63%
digits (parity)	real	400	64	320	80	50%
Gaussian (linear)	linear	400	10	320	80	48%
two-scale moons	multiscale	400	2	320	80	50%
two-scale checkerboard	multiscale	400	2	320	80	52%

Table 2: Data sets. d is the dimension after preprocessing (digits: 16 principal components). All inputs are standardised on the training fold.

one (at $k = 20$: 0.1070 against 0.0503; in excess risk, which the middle panel of Figure 1 plots, 0.0570 against 0.0003, more than a hundred times larger), the kNN-localised linear SVM in $d = 2$ is within simulation error of the global linear SVM for $k \geq 20$, the learning curve of the global linear SVM in $d = 10$ is decreasing at every step (the monotonicity that Proposition 3.2 needs), and accordingly the partition-localised linear SVM with label-independent cells grows with M (0.0577, 0.0658, 0.0803, 0.0985 for $M = 1, 2, 4, 8$).

4 Experimental protocol and predefined criterion

Data sets. Six binary tasks (Table 2), grouped into three named regimes fixed before running: *real* (wine, class 0 against the rest; breast cancer; digits, even against odd, a task with several modes per class), *linear* (the Gaussian location model of Section 3 in $d = 10$, $\delta = 3.29$, Bayes error 0.0500, so that the linear reference contains the Bayes rule), and *multiscale*, designed to favour local adaptation: two-scale moons (100 points of the standard moons and 300 points of a copy shrunk by 4 and displaced, so that scale and density both vary) and a two-scale checkerboard (100 points on 2×2 cells of side 1 and 300 points on 4×4 cells of side $\frac{1}{2}$). Breast cancer and digits are subsampled (stratified) to 300 and 400 points and digits are reduced to 16 principal components fitted on the training fold; this subsampling is part of the compute budget and is stated as such.

Resampling and tuning. Main condition: 2 repetitions of stratified 5-fold cross-validation (10 outer folds). Two robustness conditions with 5 outer folds each, on a partition different from the main one: *noise20*, in which each *training* label of a fold is flipped independently with probability 0.2 (test labels clean), and *sub25*, in which only 25% of the training fold (stratified, at least 40 points) is used. Every family is tuned on the training fold by an inner stratified 3-fold cross-validation with accuracy as the score. Grids: linear SVM $C \in \{0.001, 0.01, 0.1, 1, 10\}$; RBF-SVM $C \in \{0.1, 1, 10, 100, 1000\}$, $\gamma \in \{0.1, 0.3, 1, 3, 10, 30, 100\} \times \gamma_{\text{med}}$ with γ_{med} the inverse median squared distance; kNN-SVM $k \in \{5, 10, 20, 50, 100, \text{all}\}$ with $C = 1$ fixed, the inner validation sets being subsampled to at most 40 points to bound the number of per-point fits; cell-SVM $M \in \{1, 2, 4, 8, 16, 32\}$, $C \in \{0.1, 1, 10\}$; VB-RBF-SVM the RBF grid times $\beta \in \{0, 0.5, 1\}$ with $\kappa = 10$; LLSVM $M \in \{1, 2, 4, 8, 16, 32\}$, $C \in \{0.01, 0.1, 1, 10\}$ (values of M exceeding the number of points of an inner training fold are dropped in that fold, which only happens under sub25). Ties are broken towards the global member (Remark 2.3). These grids are those of v0.2. In the first run (v0.1) they were $C \leq 10$, $\gamma \leq 10 \gamma_{\text{med}}$, $M \leq 8$ and $k \geq 20$, and the inner cross-validation selected an edge value in most folds of the data sets that decide the result: in the 20 multiscale folds the RBF reference chose the largest γ 20 times and the largest C 20 times, cell-SVM and LLSVM the largest M 14 and 16 times, and kNN-SVM the smallest k 15 times. The grids were therefore widened at those ends (but not the C grids of cell-SVM and LLSVM; Section 5.1), with the record described below, and the selections at an edge are reported (Table 4).

Reference and paired comparison. The *best global reference* in an outer fold is whichever of the linear and RBF-SVM has the higher inner-cross-validation score in that fold, the RBF-SVM only if its score is strictly higher (ties go to the linear SVM); its test accuracy is the reference value, so the choice uses no test data. For each family and data set we record the paired difference (family minus reference) in each outer fold and a percentile bootstrap 95% interval of the mean difference over folds ($B = 2000$) [11]; an interval whose endpoint is exactly zero counts as including zero. Outer folds of a cross-validation are not independent, so these intervals are descriptive rather than exact [10, 15], and with 6 data sets no formal multi-data-set test is attempted [8]; with 10 folds they are also wide, which works against detecting small effects in either direction. For a cell with no true difference a nominal 95% interval lies entirely above zero with probability 0.025, which would give 0.6 of the 24 main-condition intervals and 1.2 of the 48 robustness ones above zero by chance (and as many below). The percentile bootstrap over so few folds is anti-conservative, however: for *independent* normal fold differences its simulated coverage is 89.9% with 10 folds and 83.5% with 5 (5.1% and 8.2% per side; `experiments/bootstrap_coverage.py`), and the dependence between folds lowers it further. We therefore read about 1.2 (main) and 4.0 (robustness) intervals above zero as what chance alone produces if every cell is null—indicative figures, since fold dependence raises them and cells with a true difference do not contribute. The majority clause of the criterion below is meant to absorb this; the single-data-set regime clause is not.

Predefined criterion of added value. A family has *added value* if (A) its mean paired improvement over the best global reference is positive with a bootstrap interval excluding zero on a majority of the 6 data sets in the main condition; or, failing that, (B) there is a named regime on every data set of which (A) holds, and no data set outside the regime shows an interval entirely below zero. Robustness is reported separately as the accuracy under the two conditions and the paired differences there. The criterion, the regimes, the families, the seed and the v0.1 grids were fixed in the code before the first run, but not in a dated record independent of the code, so that statement rests on the authors’ word; the v0.2 widening of the grids and what was to be done with the new results were written down in `PREREGISTRO_SVMF001.md` at 04:00 UTC on 3 October 2026, before the code change (04:01) and the re-run (04:04–04:08 UTC). That file was first committed together with the results and its file date is later than the run (post-run annotations were appended to it), so its timing rests on the session log of the drafting agent; the file records the SHA-256 digest of its pre-run part (its first 40 lines) and how to check it. One sensitivity check added after seeing the v0.1 results (the oracle reference of Section 5) is labelled as post hoc.

5 Results

5.1 Main condition

Table 3 gives the mean accuracies of the two references and of the selected reference together with the paired differences of the four families against the selected reference; Figure 2 shows all three conditions, and `results/tables.md` has the accuracies of every method. Of the 24 data-set/family cells, 9 have intervals entirely below zero, 1 entirely above (VB-RBF-SVM on Gaussian (linear)), and the rest include zero.

Local linear families. The three local *linear* families (kNN-SVM, cell-SVM, LLSVM) are never significantly better than the selected reference. They are below it wherever the RBF-SVM is much better than the linear SVM: on digits (cell-SVM -2.00 , LLSVM -1.13 ; kNN-SVM $+0.75$, not significant) and on 5 of the 6 multiscale cells, by 3.5 to 7.6 points (kNN-SVM ties the reference on the two-scale moons, -0.37 [-1.63 , $+1.00$]). Under the v0.1 grids the same

data set	accuracy (%)			paired difference vs. selected reference (pp), mean and 95% interval			
	linear	RBF	selected	kNN-SVM	cell-SVM	VB-RBF-SVM	LLSVM
wine	99.2	99.7	99.7	-0.28 [-1.11, +0.56]	-1.42 [-2.83, -0.01]	+0.00 [+0.00, +0.00]	-0.57 [-1.67, +0.54]
breast cancer	96.0	96.0	96.5	-1.33 [-3.34, +0.67]	-0.33 [-1.33, +0.50]	-0.50 [-1.33, +0.00]	-0.33 [-1.50, +0.67]
digits	85.2	97.5	97.5	+0.75 [-0.50, +2.00]	-2.00 [-3.50, -0.37]	+0.13 [-0.37, +0.75]	-1.13 [-2.25, -0.13]
Gaussian	93.9	93.5	93.3	-2.62 [-3.88, -1.62]	-0.25 [-1.12, +0.50]	+0.37 [+0.12, +0.75]	-0.12 [-0.87, +0.63]
moons	84.5	97.0	97.0	-0.37 [-1.63, +1.00]	-3.75 [-5.88, -1.25]	+0.12 [-1.25, +1.37]	-7.38 [-9.13, -5.75]
checkerboard	50.8	89.4	89.4	-3.50 [-6.38, -0.50]	-7.63 [-11.38, -3.25]	+0.50 [-0.50, +2.12]	-4.88 [-7.63, -1.87]

Table 3: Main condition, 10 outer folds. “Selected” is the best global reference chosen by inner cross-validation in each fold; it can be below both references’ means because it is a per-fold choice. Differences are family minus selected reference, in percentage points, with the percentile bootstrap 95% interval beneath; bold marks an interval that excludes zero.

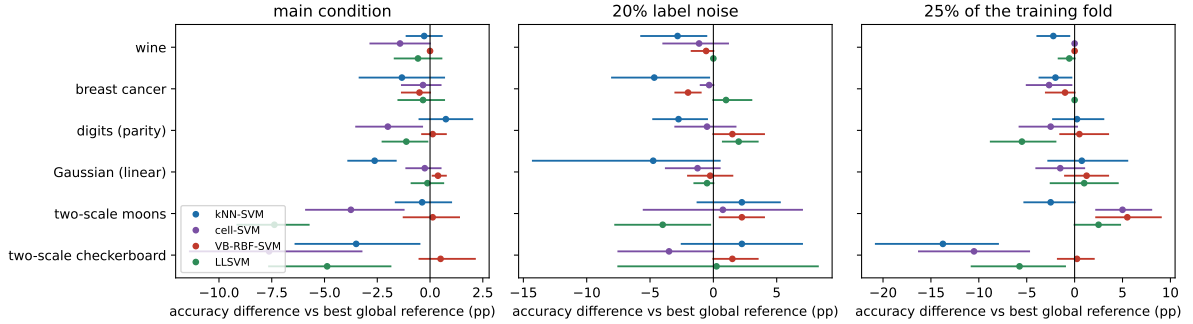


Figure 2: Paired accuracy differences against the selected best global reference (percentage points) with bootstrap 95% intervals, for the main condition and the two robustness conditions.

six cells read -10.6 to -18.2 points: most of that deficit was the grid. The widened grids still bind (Table 4; an edge was selected in at least half of the folds in 21 of its 108 data-set/edge cells). In the 20 multiscale folds kNN-SVM selects $k = 5$ in 20; LLSVM selects the largest M and the largest C together in 18, and cell-SVM in 11 (the largest $C = 10$ alone in 20). The C grids of these two families were not widened in v0.2, although v0.1 already saturated them (17 and 17 of the multiscale folds), an omission of our preregistration. The inner cross-validation thus asks for the most local configurations available, and the deficit is a statement about these grids: what the three families do not do under the grids of Section 4, which bind on both of their hyper-parameters on these sets, is reach the RBF reference (kNN-SVM ties it on the moons). Against their own global member, the linear SVM, they are far above wherever the boundary is curved (Table 7): they do adapt, reducing the approximation term that Lemma 3.1 leaves open when the reference class does not contain the Bayes rule.

The variable-bandwidth family. VB-RBF-SVM, which contains the RBF-SVM, is indistinguishable from the selected reference on the three real data sets and on the two multiscale sets for which it was designed ($+0.12$ $[-1.25, +1.37]$ and $+0.50$ $[-0.50, +2.12]$). This tie is grid-limited too. The RBF reference no longer sits at the top of its γ grid (0 of 20 multiscale folds, against 20 of 20 in v0.1) and is at the top of its C grid in 6; but VB-RBF-SVM selects the largest $\gamma = 100\gamma_{\text{med}}$ in 6 of the 10 moons folds, and the largest $C = 1000$ is selected

method: hyper-parameter	wine	breast c.	digits	Gaussian	moons	checkerb.	all
linear SVM: C	0/0	0/1	0/2	0/1	0/3	7/1	7/8
RBF-SVM: γ	2/0	5/0	0/0	8/0	0/0	0/0	15/0
RBF-SVM: C	0/0	0/0	0/0	0/0	0/1	0/5	0/6
kNN-SVM: k	1/-	2/-	8/-	0/-	10/-	10/-	31/-
cell-SVM: M	-/1	-/0	-/6	-/0	-/4	-/7	-/18
cell-SVM: C	7/0	3/0	3/0	8/2	0/10	0/10	21/22
VB-RBF-SVM: γ	2/0	4/0	0/0	7/0	0/6	0/0	13/6
VB-RBF-SVM: C	0/0	0/0	0/0	0/0	0/0	0/4	0/4
VB-RBF-SVM: β	-/0	-/0	-/4	-/2	-/2	-/3	-/11
LLSVM: M	-/0	-/1	-/7	-/1	-/9	-/10	-/28
LLSVM: C	0/0	0/2	0/10	4/1	0/9	0/10	4/32

Table 4: Grid saturation in the main condition: number of outer folds (of 10 per data set, 60 in all) in which the inner cross-validation selected the lowest/highest value of the widened grid (lower/upper, “-” when that edge is the family’s global member and is not a saturation: $k = \text{all}$, $M = 1$, $\beta = 0$). For M the top of the grid available in the fold is used.

in 4 (VB-RBF-SVM) and 5 (RBF-SVM) of the 10 checkerboard folds. With larger grids the comparison could move in either direction. On the Gaussian linear data set, where Section 3 says that local adaptation cannot reduce the excess risk, the family shows $+0.37$ [$+0.12$, $+0.75$]. This cell amounts to 3 test points out of 800: in 2 folds VB-RBF-SVM ($\beta > 0$) beat its own RBF member by one point, and in 1 fold the inner cross-validation chose the linear SVM, one point worse than the RBF. The selected reference (93.3) is below the mean of the linear SVM (93.9) because of one fold in which the RBF was chosen and lost 5 points to it (the other differences cancel; of the 8 folds in which the RBF was chosen, the linear SVM was better in 2, equal in 5 and worse in 1), and below that of the RBF-SVM (93.5) because of 2 folds in which the linear SVM was chosen and lost one point each. Against the family’s own fixed member, the RBF-SVM with the same grid, the Gaussian difference is $+0.12$ [-0.25 , $+0.50$], and VB-RBF-SVM is above its own member on 0 of 6 data sets (Table 7). Against the post-hoc *oracle* reference, the better of the two global references on each test fold (optimistic, since it uses test data; defined after the v0.1 results were seen), the Gaussian cell is -0.62 [-2.00 , $+0.25$] and no main-condition interval is above zero (0 of 24; `results/posthoc_oracle.md`). The inner cross-validation did select genuinely local configurations: $\beta > 0$ in 60% of the Gaussian folds and in 100% and 50% of the multiscale folds, and $M > 1$ or $k \neq \text{all}$ in most folds of the data sets where the local linear families then lose (`results/tables.md`).

5.2 Robustness: label noise and subsampling

Averaged over the data sets, the selected reference loses 7.1 points under label noise and 6.4 under subsampling (all accuracies in `results/tables.md`). These drops are not comparable across families, which start from different accuracies, and the robustness conditions use another partition into folds; only the paired comparison within each condition (Figure 2) is used. There, 4 of the 48 intervals lie above zero (LLSVM on digits (parity) under noise; VB-RBF-SVM on two-scale moons under noise; cell-SVM on two-scale moons under sub; VB-RBF-SVM on two-scale moons under sub), against up to about 4.0 expected by chance (Section 4), and 12 below. 3 of the positives persist against both the fixed RBF-SVM and the post-hoc oracle reference of Section 5.1, all on the two-scale moons: VB-RBF-SVM under noise ($+2.25$ [$+0.50$, $+4.00$]) and under subsampling ($+5.50$ [$+2.25$, $+9.00$]), and cell-SVM under subsampling, with 80 training points ($+5.00$ [$+2.25$, $+8.00$]; $+4.75$ [$+2.00$, $+7.25$] against the oracle); the cell of LLSVM on digits (parity) under noise does not (`results/posthoc_oracle.md`). With 5 folds, no predefined criterion for robustness and a count compatible with chance, these are hypotheses

family	CI > 0	CI < 0	majority (A)	named regime (B)	added value
kNN-SVM	0/6	2/6	not met	none	no
cell-SVM	0/6	4/6	not met	none	no
VB-RBF-SVM	1/6	0/6	not met	linear	yes (by the letter; see text)
LLSVM	0/6	3/6	not met	none	no

Table 5: The predefined criterion applied to the main condition (6 data sets).

for a larger run, not results. The kNN-SVM is below the reference on 3 of 6 data sets under noise and 3 under subsampling (2 in the main condition), the only robustness pattern with a consistent sign; it is consistent with, though not covered by, the variance mechanism of Section 3. We claim no superior robustness for any family.

5.3 Verdict under the predefined criterion

Table 5 applies the criterion of Section 4 literally. No family meets the majority clause (A). The regime clause (B) is met, by the letter, by 1 family in one regime: VB-RBF-SVM in the *linear* regime, which consists of a single data set (kNN-SVM: none; cell-SVM: none; LLSVM: none). We report it as the rule requires and say why it is not added value in the sense intended: the regime is the one in which Section 3 shows that no reduction of excess risk is available, the gain is against the selected reference and vanishes against the family’s own global member and against the oracle reference, a one-data-set regime is too weak a base for a named-regime claim, and a single interval above zero among 24 is what chance alone would produce (about 1.2 expected, Section 4). The flaw is in our criterion, which should have required at least two data sets per regime; we record it rather than repair it after the fact. In the multiscale regime, for which the variable-bandwidth family was designed, the outcome is a tie with the RBF-SVM under grids that still bind on both (Section 5.1).

6 What can be claimed

Limitations. The families are reconstructions chosen for transparency, not the strongest versions in the literature (linear local SVMs, kNN-SVM with a fixed C and a subsampled inner validation, partitions from a single k -means run, three values of the bandwidth exponent). The data sets are small by design (about 259 CPU-seconds in all), so the comparison speaks to a few hundred training points, where Section 3 shows a variance cost for label-independent partitions; on the multiscale sets the inner cross-validation nonetheless asked for the most local configurations available, so the deficit there is not attributed to that mechanism. The grids are finite and still bind where Table 4 shows edge selections—including the C grids of cell-SVM and LLSVM, not widened although v0.1 had saturated them, and the γ grid of VB-RBF-SVM on the moons—and the comparisons there are grid-limited. The bootstrap intervals treat dependent folds as exchangeable and under-cover (Section 4); no multiplicity correction is applied beyond the majority clause. Proposition 3.2 covers label-independent partitions only. The regime clause admitted a single-data-set regime, a design flaw. The predefinition of the v0.1 protocol is not independently verifiable, and the timing of the v0.2 preregistration rests on a session log (Section 4).

Next steps for the research line. (a) A lower bound for neighbourhood-based localisation: for the kNN-localised linear rule in the Gaussian model, bound the conditional excess risk at a query by the local sample size, which would extend Proposition 3.2 to the kNN-SVM. (b) Re-run the comparison at n in the thousands with the same criterion (repaired to require two data sets per regime), with γ above $100\gamma_{\text{med}}$, C up to 1000 for cell-SVM and LLSVM and

claim	status
Localisation acts only on the estimation term if the reference class contains a Bayes rule (Lemma 3.1); with label-independent cells the risk is the learning curve at binomially thinned sizes, not below the global rule if the curve is non-increasing, strictly above if $M \geq 2$ and it decreases at the last step (Prop. 3.2); K_a is a positive-definite kernel nesting the RBF kernel (Lemma 2.2)	proved here
Gaussian model: closed-form learning curve of the nearest-centroid threshold, strictly decreasing for $n \geq 1$; four label-independent cells at $n = 320$ multiply its excess risk by 4.08 (Prop. 3.4, Cor. 3.5)	proved; exact numbers
Linear SVM learning curve decreasing in that model ($d = 10$); partition-localised SVM risk grows with M ; kNN-localised rules do not improve on the global rule there (Appendix B)	verified, not proved
No family meets the criterion; the 1 interval above zero is against the selected reference, compatible with chance, and absent against the family's own global member	measured
Local linear families are below the RBF reference on digits and on 5 of 6 multiscale cells, by 3.5 to 7.6 points (kNN-SVM ties it on the moons), with the inner cross-validation at the local edge of both of their hyper-parameters	measured, grid-limited
VB-RBF-SVM ties the RBF-SVM on the multiscale sets, with its own γ at the top of the grid in 6 of 10 moons folds and C at the top for both in 4–5 of 10 checkerboard folds	measured, grid-limited
Under label noise and subsampling 4 of 48 intervals are above zero (up to about 4.0 expected by chance; 3 on the moons persist against the oracle); kNN-SVM is below the reference on 3 and 3 of 6 data sets	measured; the positives are hypotheses
General superiority of any family in accuracy or robustness; the original variants, larger samples, other kernels	not claimed

Table 6: Claims and their status. “Verified” and “measured” mean produced by the scripts of Appendix A with the fixed seed; neither is a proof.

a criterion for robustness, all in a preregistration committed before any code change, to see whether the multiscale tie turns into a gain or a loss for the variable-bandwidth family. (c) Compare the variable-bandwidth kernel with the simpler alternative of a global RBF-SVM on density-normalised inputs, which may capture the same effect without the dimension-dependent prefactor of Lemma 2.2. (d) If the original variants of the line are recovered, insert them into this protocol unchanged.

A Reproducibility

The directory contains `experiments/families.py` (the six families in about 300 lines), `experiments/run_comparison.py` (protocol, criterion, bootstrap, grid-saturation counts, `results/results.json`, `results/tables.md`, Figure 2), `experiments/exact_example.py` (Section 3: exact sums, Monte Carlo, D1–D4, `results/exact_example.json`, Figure 1), `experiments/posthoc_oracle.py` (oracle and own-member comparisons, from `results.json` only), `experiments/make_region_figure.py` (Figure 3) and `experiments/make_numbers.py`, which converts the JSON results into the macros and table bodies used by this document, so that no number in the text is typed. Seed 20260930; Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, scikit-learn 1.9.1 [16]; single-threaded BLAS; 231 seconds for the comparison and 28 seconds for the exact example on a shared four-core machine. `run_comparison.py` with `-fast` runs a reduced smoke test; the v0.1 results are kept in `results/v01/`. Implementation details are in the README. One detail matters for Lemma 2.2: the implementation computes r_κ excluding the point itself for training points and not for queries, so the bandwidth map is not a single

data set	kNN-SVM vs. linear SVM	cell-SVM vs. linear SVM	VB-RBF-SVM vs. RBF-SVM	LLSVM vs. linear SVM
wine	+0.29 [-0.56, +1.15]	-0.85 [-1.70, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
breast cancer	-0.83 [-2.33, +0.50]	+0.17 [-0.33, +0.67]	+0.00 [+0.00, +0.00]	+0.17 [-1.00, +1.17]
digits (parity)	+13.00 [+11.50, +14.50]	+10.25 [+8.12, +12.25]	+0.13 [-0.37, +0.75]	+11.12 [+8.87, +13.12]
Gaussian (linear)	-3.25 [-5.12, -1.63]	-0.87 [-2.12, +0.25]	+0.12 [-0.25, +0.50]	-0.75 [-1.87, +0.25]
two-scale moons	+12.12 [+10.50, +13.62]	+8.75 [+6.00, +11.50]	+0.12 [-1.25, +1.38]	+5.12 [+3.75, +6.38]
two-scale checkerboard	+35.12 [+32.25, +37.88]	+31.00 [+27.38, +34.38]	+0.50 [-0.50, +2.12]	+33.75 [+31.00, +36.38]

Table 7: Nested comparison, main condition: each family against its own fixed global reference (the tuned linear SVM for the three local linear families, the tuned RBF-SVM for VB-RBF-SVM), percentage points with 95% intervals, bold when the interval excludes zero. For kNN-SVM the member actually contained in its grid is the linear SVM with $C = 1$ fixed, so its column is an approximation of the nested comparison. 9 intervals are above zero and 1 below.

function of x at the training points and $K_a(z, x_i) \not\rightarrow 1$ as $z \rightarrow x_i$; the training Gram matrix is K_a for the map restricted to the training set, so its positive semidefiniteness and all reported numbers are unaffected.

B Simulations for rules not covered by Proposition 3.2

The middle and right panels of Figure 1 report simulations in the model of Proposition 3.4 (computationally verified, not proved); the full tables, with standard errors, are in `results/exact_example.md`. D1 uses 200 training samples and the exact conditional risk averaged over 1000 test points; D2 16 samples and 125 test points with $C = 1$ (its “global” value is the exact risk of the fitted hyperplane and its kNN values the exact conditional risk averaged over the test points, two different estimators of the same quantity); D3 200 samples per n ; D4 200 samples with cells drawn independently of the data. (D1) The kNN-localised nearest-centroid rule in $d = 2$ (second coordinate pure noise), $n = 320$: the risk is worst at $k = 20$ (0.1070, against 0.0503 for $k = n$), because the local class means are truncated by the neighbourhood (bias) and computed from few points (variance); for small k the neighbourhoods are mostly pure and the rule degenerates to a k -NN vote [7, 20]. (D2) The kNN-localised linear SVM in $d = 2$ with $C = 1$: 0.0562 at $k = 10$ and 0.0540 at $k = 160$ against 0.0521 ± 0.0006 for the global linear SVM; the differences at $k \geq 20$ are about one simulation standard error, so in two dimensions the localised SVM neither helps nor clearly hurts here. (D3) The learning curve of the global linear SVM in $d = 10$ is decreasing at every step between $n = 20$ (0.1178) and $n = 640$ (0.0538). (D4) The partition-localised linear SVM with label-independent cells in $d = 10$, $n = 320$, has risk 0.0577, 0.0658, 0.0803, 0.0985 for $M = 1, 2, 4, 8$.

C Nested comparison and decision regions

The mean accuracy of every method in every condition, the per-data-set paired differences of the two robustness conditions, the fraction of folds in which a local configuration was selected, and the grid-saturation counts of the robustness conditions are in `results/tables.md`; the post-hoc comparisons against the oracle reference are in `results/posthoc_oracle.md`. Table 7 and Figure 3 are kept here because the text relies on them.

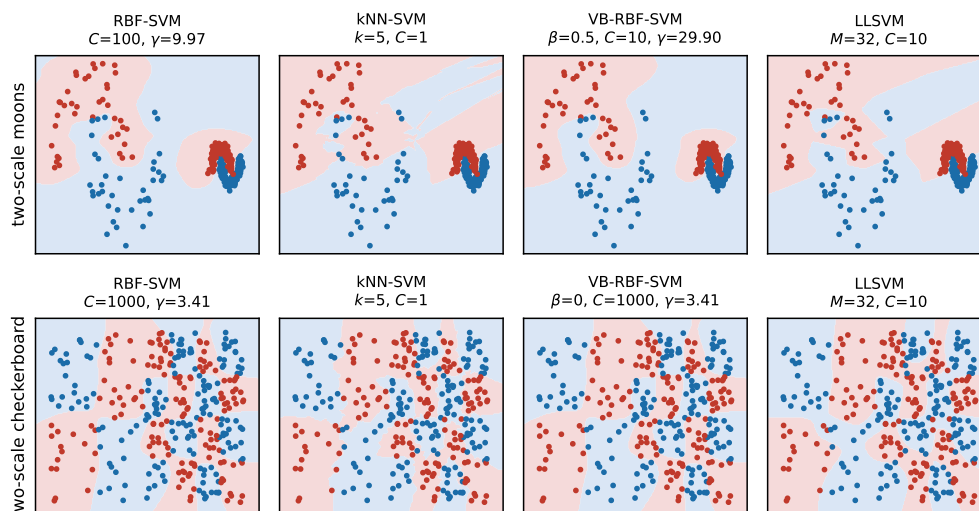


Figure 3: Decision regions on the first outer training fold of the two multiscale data sets, for the configurations selected by inner cross-validation in that fold (given in the panel titles).

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